$$\nabla_X = \frac{\partial}{\partial X} = \begin{bmatrix} \frac{\partial}{\partial x_{11}} & \cdots & \frac{\partial}{\partial x_{1n}} \\ \vdots & \vdots & \vdots \\ \frac{\partial}{\partial x_{m1}} & \cdots & \frac{\partial}{\partial x_{mn}} \end{bmatrix} \qquad \dots (3.20)$$

The gradient vectors of functions $f(x)$ and $f(X)$ are defined respectively as

$$\nabla_X f(X) = \begin{bmatrix} \frac{\partial f(X)}{\partial x_1} & \cdots & \frac{\partial f(x)}{\partial x_m} \end{bmatrix}^T = \frac{\partial f(X)}{\partial X}$$

$$\nabla_{vec(X)} f(X) = \begin{bmatrix} \frac{\partial f(X)}{\partial x_{11}} & \cdots & \frac{\partial f(X)}{\partial x_{m1}} & \cdots & \frac{\partial f}{\partial x_{1n}} & \cdots & \frac{\partial f(X)}{\partial x_{mn}} \end{bmatrix}^T$$

The Gradient matrix of the function f(x) is defined as

$$\nabla_X f(X) = \begin{bmatrix} \frac{\partial f(X)}{\partial x_{11}} & \cdots & \frac{\partial f(X)}{\partial x_{1n}} \\ \vdots & \vdots & \vdots \\ \frac{\partial f(X)}{\partial x_{m1}} & \cdots & \frac{\partial f(x)}{\partial x_{mn}} \end{bmatrix} = \frac{\partial f(X)}{\partial X} \qquad \dots (3.21)$$

Clearly, there is a relation between gradient matrix and gradient vector of a function $f(X)$.

$$vec\nabla_X f(X) = \nabla_{vecX} f(X) \qquad \dots (3.22)$$

Further, we can write

$$\nabla_X f(X) = D_X^T f(X)$$

that is, the gradient matrix of a real scalar function $f(X)$ is equal to the transpose of its Jacobian matrix.

Example 3.11:

Let $f(X): R^3 \to R$ be defined as $f(X) = x_1^2 + 2x_1x_2 + 3x_2^2 + 4x_1x_2x_3 + x_3^3$ for

$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. Calculate $\nabla_X f(X)$.

The solution is

$$\nabla_X f(X) = \begin{bmatrix} \frac{\partial f(X)}{\partial x} \\ \frac{\partial f(X)}{\partial x_2} \\ \frac{\partial f(X)}{\partial x_3} \end{bmatrix} = \begin{bmatrix} 2x_1 + 2x_2 + 4x_2x_3 \\ 2x_1 + 6x_2 + 4x_1x_3 \\ 4x_1x_2 + 3x_3^2 \end{bmatrix}$$

3.11.3 Real Hessian Matrix

Let us Consider a real vector $X \in \mathbb{R}^{m\times 1}$, the second-order derivative of a real function $f(X)$ is known as the Hessian matrix, denoted by H$[f(X)]$ and is defined as

$$H[f(X)] = \frac{\partial^2 f(X)}{\partial X \partial X^T} = \frac{\partial}{\partial X}\left(\frac{\partial f(X)}{\partial X^T}\right) \in \mathbb{R}^{m\times m}$$

which is simply represented as

$$H[f(X)] = \nabla_X^2 f(X) = \nabla_X(D_X f(X)) \qquad \dots (3.23)$$

In words, Hessian matrix is the gradient matrix of the Jacobian matrix of the function $f(X)$ with respect to *X*.

Hence the (i, j)th entry of the Hessian matrix is defined as

$$[Hf(X)]_{ij} = \left[\frac{\partial^2 f(X)}{\partial X \partial X^T}\right]_{ij} = \frac{\partial}{\partial x_i}\left(\frac{\partial f(X)}{\partial x_j}\right)$$

Therefore, the Hessian matrix is

$$H[f(X)] = \frac{\partial^2 f(X)}{\partial X \partial X^T} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_1 x_m} \\ \vdots & \vdots & \vdots \\ \frac{\partial^2 f}{\partial x_m \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_m \partial x_n} \end{bmatrix} \in \mathbb{R}^{m\times m} \qquad \dots (3.24)$$

Equation (3.24) shows that the Hessian matrix of a real-valued scalar function $f(X)$ is an m×m matrix that consists of m^2 second-order partial derivatives of $f(X)$ with respect to the entries x_i of the vector variable *X*.

By definition, we can show that the Hessian matrix of a real scalar function $f(X)$ is a real symmetric matrix, i.e.,

$$(H[f(x)])^T = H[f(X)]$$

Because $f(X)$is a second order differentiable continuous function, its second-order derivative is independent of the order of differentiation. It means,

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i} \qquad \dots (3.25)$$

Example 3.12:

Find Hessian matrix of the following function :

$f(X) = x_1^2 + 2x_1x_2 + 3x_2^2 + 4x_1x_2x_3 + x_3^3$.

The Hessian matrix is given by

$$\nabla_X^2 f(X) = \begin{bmatrix} \frac{\partial f}{\partial x_1^2} & \frac{\partial x^2 f}{\partial x_1 x_2} & \frac{\partial f^2}{\partial x_1 \partial x_3} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \frac{\partial^2 f}{\partial x_2 \partial x_3} \\ \frac{\partial^2 f}{\partial x_3 \partial x_1} & \frac{\partial x^2 f}{\partial x_3 \partial x_2} & \frac{\partial^2 f}{\partial x_3} \end{bmatrix} \in R^{3\times 3}$$

$$H[F(X) = \begin{bmatrix} 2 & 2+4x_3 & 4x_2 \\ 2+4x_3 & 6 & 4x_1 \\ 4x_2 & 4x_1 & 6x_3 \end{bmatrix}_{3\times 3}$$

Note that the Hessian matrix is symmetric.

Check Your Progress 3

1) Given the fact that for any two vectors, **a** and **b**, c $vec\ \boldsymbol{ab}' = \boldsymbol{b} \otimes \boldsymbol{a}$. Show that $vec\ \boldsymbol{ABC} = (\boldsymbol{C}' \otimes \boldsymbol{A})vecB$, whenever the product **ABC** is defined.

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2) Let $x_1, x_2 \ldots x_n$ be an independent and identically distributed (iid) sample from $N(\mu, \sigma^2)$. Let $\theta = (\mu, \sigma^2)$. If log-likelihood function $log\, L\,(\theta) = -\frac{n}{2} ln\, \sigma^2 - \frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2$, then derive the Fisher information.

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3.12 LET US SUM UP

In this Unit we focussed on multiplication and inversion of matrices. For invertibility, a square matrix must be full rank. Therefore, we discussed about the method of finding the rank of a matrix by the reduced echelon form method. A matrix of large dimension can be partitioned into various blocks, which make calculation easy. Thus, we presented the results of matrix operations of a partitioned matrix. Diagonalization of matrix and eigenvalue of matrix are also discussed. We discussed certain special matrices which are useful in econometrics. In the end, we discussed Kronecker product, vectorization of matrix and matrix differentiation.

3.13 KEYWORDS

Linear Independence of Vectors : A set of n vectors, denoted $\{U_1, U_2, \dots U_n\}$ is said to be linearly independent if the matrix equation

$c_1u_1 + \dots + c_nu_n = 0$

has only zero solution $c_1 \dots = c_n = 0$. If the above equation holds for a set of coefficient that are not all zero, then the n vectors are said to be linearliy independent.

Euclidean Norm or $\boldsymbol{l_2}$ **Norm of a Vector** : $\|\boldsymbol{x_2}\|_2 = \|\boldsymbol{X}\|_E = \left(|\boldsymbol{x}_1^2| + |\boldsymbol{x}_2|^2 \dots\right)^2$. A vector with unit Euclidean length is referred to as a normalized vector.

Idempotent Matrix : A matrix $\boldsymbol{A}_{n\times n}$ is idempotent if $\boldsymbol{A}^2 = \boldsymbol{AA} = \boldsymbol{A}$. Its eigenvalues are 0 and 1 except for identity matrix but converse is not necessarily true.

Characteristic Polynomial : $P_{nx}{}^n + P_{n-1}x^{n-1} \dots . P_{1x} + P_0$ is known as the characteristic polynomial of A and $P(x) = det(A - xI) = 0$ is said to be characteristic equation of A.

3.14 SOME USEFUL BOOKS

Abadir. M. Karim and Magnus. R. Jan (2005), *Matrix Algebra*, Cambridge University Press.

Gentle. E. James, 2017, *Matrix Algebra: Theory, Computations and Applications in Statistics*, Second Edition, Springer International Publishing AG.

Magnus, R. Jan, and Neudeckery Heinz, 2019, *Matrix Differential Calculus with Applications in Statistics and Econometrics*, Third Edition, John Wiley & Sons.

Schoot, R. James, 2017, *Matrix Analysis for Statistics*, Third Edition, John Wiley & Sons.

Turking-ton, A Darrell, 2005, *Matrix Calculus and Zero-One Matrices: Statistical and Econometric Applications*, Cambridge University Press.

3.15 ANSWERS/HINT TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) We can rewrite both the equations with endogenous variables C and Y on the left side

$Y - C = I + G$

$-bY + C = a$

Writing above equations in matrix notation, we obtain

$$\begin{bmatrix} 1 & -1 \\ -b & 1 \end{bmatrix}\begin{bmatrix} Y \\ C \end{bmatrix} = \begin{bmatrix} I + G \\ a \end{bmatrix}$$

This resembles the equation **AX = B** where $\boldsymbol{X} = \begin{bmatrix} Y \\ C \end{bmatrix}$

Hence, $\begin{bmatrix} Y \\ C \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -b & 1 \end{bmatrix}^{-1} \begin{bmatrix} I + G \\ a \end{bmatrix}$

$$\Rightarrow X = \begin{bmatrix} Y \\ C \end{bmatrix} = \frac{-1}{1-b}\begin{bmatrix} 1 & 1 \\ b & 1 \end{bmatrix}\begin{bmatrix} I + G \\ a \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} Y \\ C \end{bmatrix} = \begin{bmatrix} \frac{I+G+a}{1-b} \\ \frac{(1+G)b+a}{1-b} \end{bmatrix}.$$

2) $\boldsymbol{A} = \begin{bmatrix} 2 & 4 & 1 \\ 4 & 3 & 7 \\ 2 & 1 & 3 \end{bmatrix}$

First, compute the cofactor of C_{11}. For this, you have to delete row 1 and column 1 of matrix **A**. Next, compute determinant of the residual 2 × 2 matrix.

Thus, $C_{11} = (-1)^{1+1}\begin{vmatrix} 3 & 7 \\ 1 & 3 \end{vmatrix} = 2.$

Similarly, you should compute

$C_{12} = (-1)^{1+2}\begin{vmatrix} 4 & 7 \\ 2 & 3 \end{vmatrix} = 2$

$C_{13} = (-1)^{1+3}\begin{vmatrix} 4 & 3 \\ 2 & 1 \end{vmatrix} = -2$

$C_{21} = (-1)^{2+1}\begin{vmatrix} 4 & 1 \\ 1 & 3 \end{vmatrix} = -11$

$C_{22} = (-1)^{2+2}\begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix} = 4$

$C_{23} = (-1)^{2+3}\begin{vmatrix} 2 & 4 \\ 2 & 1 \end{vmatrix} = 6$

$C_{31} = (-1)^{3+1}\begin{vmatrix} 4 & 1 \\ 3 & 7 \end{vmatrix} = 25$

$C_{32} = (-1)^{3+2}\begin{vmatrix} 2 & 1 \\ 4 & 7 \end{vmatrix} = -10$

$C_{33} = (-1)^{3+3} \begin{vmatrix} 2 & 4 \\ 4 & 3 \end{vmatrix} = -10$

The cofactor matrix $= \begin{bmatrix} 2 & 2 & -2 \\ -11 & 4 & 6 \\ 25 & -10 & -10 \end{bmatrix}$

$det\, A = a_{21}.C_{21} \quad + \quad a_{22}.C_{22} \quad + \quad a_{23}.C_{23}$

$\quad = (4-11) \quad + \quad (3.4) \quad + \quad 7.6 = 10$

Adjoint matrix = (cofactor matrix)^T

$= \begin{bmatrix} 2 & -11 & 25 \\ 2 & 4 & -10 \\ -2 & 6 & -10 \end{bmatrix}$= adjoint **A**

Therefore, $\boldsymbol{A^{-1}} = \frac{1}{det\, A} adjo\, int\, A = \begin{bmatrix} 115 & -\frac{11}{10} & 5/2 \\ 115 & 2/5 & -1 \\ -1/5 & 3/5 & -1 \end{bmatrix}$

3) The matrix for determination of rank is

$$\begin{pmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{pmatrix}$$

$$\xrightarrow{R_2 \to R_2 + 2R_1} \begin{pmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 21 & -21 & 0 & -15 \end{pmatrix}$$

$$\xrightarrow{R_3 \to R_3 - 7R_1} \begin{pmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & -21 & -14 & -29 \end{pmatrix}$$

$$\xrightarrow{R_3 \to R_3 + \frac{1}{2}R_2} \begin{pmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Above, we converted the original matrix to row echelon form by applying elementary row operations. The number of non-zero rows is 2; hence the rank of the matrix is 2. Recall that we can interpret rank = no of pivots. Thus, rank = no of pivots = 2.

Check Your Progress 2

1. If **A** has n linearly independent eigenvectors $x_1, x_2 \ldots . x_n$, then $\boldsymbol{T} = (x_1 \ldots x_n)$ is non-singular and satisfies

$$\boldsymbol{AT} = \boldsymbol{A}(x_1, \ldots, x_n) = (\boldsymbol{A}x_1 \ldots . \boldsymbol{A}x_n) = (\lambda_1 x_1 \ldots . \lambda_n x_n)$$

$$= (x_1, x_2 \ldots x_n)\boldsymbol{\Lambda} = \boldsymbol{T\Lambda}$$

where $\boldsymbol{\Lambda} = diag.(\lambda_1, \lambda_2 \ldots . \lambda_n)$ and $\lambda_1, \lambda_2 \ldots \lambda_n$ are eigen values of **A**. Hence $\boldsymbol{T^{-1}AT} = \boldsymbol{\Lambda}$**.** Conversely, if $\boldsymbol{AT} = \boldsymbol{T\Lambda}$ for some diagonal matrix $\boldsymbol{\Lambda}$, then $\boldsymbol{A}(\boldsymbol{T}e_i) = (\boldsymbol{AT})e_i = (\boldsymbol{T\Lambda})e_i = \boldsymbol{T}(\boldsymbol{\Lambda}e_i) = \boldsymbol{T}(\lambda_i e_i) = \lambda_i(\boldsymbol{T}e_i)$

where e_i denotes the ith unit vector, a vector which has 1 at ith place and zero in all other places.

Hence, each column of **T** is an eigenvector of **A** and since **T** is non-singular, these eigenvectors are linearly independent.

2. (a) It is given that $\boldsymbol{M} = \boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'}$

We have to show that **M** is symmetric and idempotent. First, the symmetric part. A matrix is symmetric when $\boldsymbol{A} = \boldsymbol{A'}$. By applying this argument $\boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'}$ is symmetric. Second, the idempotent part. Check whether **MM** = **M**

$$(\boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'})(\boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'})$$
$$= \boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'} + \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'}$$
$$= \boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'} + \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'}$$
$$= \boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'} = \boldsymbol{M}$$

(b) We have to verify whether $\boldsymbol{MX} = \boldsymbol{0}$.

$$\boldsymbol{MX} = (\boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'})\boldsymbol{X}$$
$$= \boldsymbol{X} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'X} = \boldsymbol{X} - \boldsymbol{X} = \boldsymbol{0}$$

(c) Using the rule, rank of an idempotent matrix is the trace of that matrix.

rank (**M**) = trace (**M**) $= tr(\boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'})$

$$= tr(\boldsymbol{I_n}) - tr(\boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'})$$
$$= tr(\boldsymbol{I_n}) - tr(\boldsymbol{X'X}(\boldsymbol{X'X})^{-1})$$
$$= n - tr(\boldsymbol{I_k}) = n - k$$

3. We know that $\boldsymbol{X'V^{-1}M} = \boldsymbol{0}$

(a) $\Rightarrow \boldsymbol{X'V^{-1}}(\boldsymbol{I_n} - \boldsymbol{X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'}) = \boldsymbol{0}$

$$\Rightarrow \boldsymbol{X'V^{-1}} = \boldsymbol{X'V^{-1}X}(\boldsymbol{X'X})^{-1}\boldsymbol{X'}$$
$$\Rightarrow (\boldsymbol{X'V^{-1}X})^{-1}\boldsymbol{X'V^{-1}} = (\boldsymbol{X'X})^{-1}\boldsymbol{X'}$$

(b) Post-multiply both sides of equality in the above by $\boldsymbol{VX}(\boldsymbol{X'X})^{-1}$. We get

$$(\boldsymbol{X'V^{-1}X})^{-1}\boldsymbol{X'V^{-1}VX}(\boldsymbol{X'X})^{-1} = (\boldsymbol{X'X})^{-1}\boldsymbol{X'VX}(\boldsymbol{X'X})^{-1}$$
$$\Rightarrow (\boldsymbol{X'V^{-1}X})^{-1} = (\boldsymbol{X'X})^{-1}\boldsymbol{X'VX}(\boldsymbol{X'X})^{-1}$$

Check Your Progress 3

1. Let $\boldsymbol{b} = (b_1 \dots . b_n)'$Then,

$$vec(\boldsymbol{ab'}) = vec(b_1 a_1 \dots b_n a) = \begin{pmatrix} b_1 a \\ \vdots \\ b_n a \end{pmatrix} = \boldsymbol{b} \otimes \boldsymbol{a}$$

Let $\boldsymbol{B} = (\boldsymbol{b_1}, \boldsymbol{b_2}, \dots, \boldsymbol{b_n})$ be a m×n matrix. Let us denote the n column of $\boldsymbol{I_n}$ matrix by $e_1 \dots e_n$. Then, **B** can be written as $\boldsymbol{B} = \sum_{i=1}^{n} b_i e_i'$

Hence, $vec(\boldsymbol{ABC}) = vec \sum_{i=1}^{n} \boldsymbol{A} b_i e_i' \boldsymbol{C} = \sum_{i=1}^{n} vec((\boldsymbol{A}b_i)(C'e_i)')$

$$= \sum_{i=1}^{n} (\boldsymbol{C'}e_i) \otimes (\boldsymbol{A}b_i) = (\boldsymbol{C'} \otimes \boldsymbol{A}) \sum_{i=1}^{n} e_i \otimes b_i$$

$= (\boldsymbol{C}' \otimes \boldsymbol{A}) \sum_{i=1}^{n} vec(b_i e_i') = (\boldsymbol{C}' \otimes A) vec(\boldsymbol{B})$

2. Let us re-state the problem: We have $x_1, x_2, \ldots, x_n$ be iid (independent and identically distributed) sample from normal distribution $N(\mu, \sigma^2)$ where μ =mean, σ^2 =variance

Let the parameter (the unknowns to be estimated) space $\theta = (\mu, \sigma^2)$

The log-likelihood function is

$$log\, L(\theta) = -\frac{n}{2} ln\, \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2$$

The score function is now a gradient vector

$$S(\theta) = \frac{\partial}{\partial \theta} log\, L(\theta) = \begin{pmatrix} \frac{\partial}{\partial \mu} ln\, L(\theta) \\ \frac{\partial}{\partial \sigma^2} ln\, L(\theta) \end{pmatrix} = \begin{pmatrix} \frac{n}{\sigma^2}(\bar{x} - \mu) \\ -\frac{n}{2\sigma^2} + \frac{\sum (x_i - \mu)^2}{2\sigma^4} \end{pmatrix}$$

The observed Fisher information is minus the Hessian of the log-likelihood function. Thus,

$$I(\theta) = -\frac{\partial^2}{\partial \theta \partial \theta'} log\, L(\theta)$$

$$I(\theta) = \begin{pmatrix} \frac{n}{\sigma^2} & \frac{n}{\sigma^4}(\bar{x} - \mu) \\ \frac{n}{\sigma^4}(\bar{x} - \mu) & -\frac{n}{2\sigma^4} + \frac{\sum (x_i - \mu)^2}{(\sigma^2)^3} \end{pmatrix}$$

The expected Fisher information is

$I(\theta) = E_\theta I(\theta)$.

The above is called the information matrix, a 2 × 2 variance matrix and it is a positive semi-definite matrix.

$$I(\theta) = \begin{pmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{pmatrix}$$

$det\, d\,(\theta)$ is always positive because n and σ^2 are always positive.